

Option decision (LSM vs market) · tables

Friday 17 March 2023 monthly expiry · 1-hour lookback · SPY / AAPL / MSFT

What this PDF is for

- This PDF records option-exercise (optimal stopping) results: Longstaff-Schwartz American call prices versus listed market quotes.
- The same 12 contracts per ticker are priced under every model, so differences come from the dynamics, not from the sample.
- Six models are in the tables, including Heston (no jumps) and Heston-Merton. On this 1-hour / 1-minute calibration both Heston and Heston-Merton risk-neutral paths often explode, so their LSM RMSE is typically ~90-100.
- The first results page is the equal-weight mean of SPY, AAPL, and MSFT. Later pages are per ticker.
- In every block the best model is bold and shaded: lowest RMSE of LSM price versus market option_price.

Methodology

- Calibration and rolling are the same as the stock-path study (1-hour lookback; hourly and minutely; 1-day and 5-day windows).
- Contracts are 12 listed American calls per name from the short-interval options panel, sampled on the study window (seed 42).
- The sample is OTM / ATM / ITM $\times$ DTE 7-60 trading days, with unique 1-minute RTH quotes.
- Each contract is priced with Longstaff-Schwartz LSM: 500 risk-neutral paths, $n_steps = DTE$, $\Delta t = 1/252$.
- RMSE and MAE are computed on (model LSM price – listed market price) across the 12 contracts.
- The same seeds and the same 12 contracts are reused across models, including Heston.
- Models priced here: GBM, Merton, Heston, Heston-Merton, GARCH(1,1), and GARCH-Merton.

Option LSM vs market · mean of SPY / AAPL / MSFT

1-day · hourly rolling

Model	RMSE	MAE
GBM	2.595	2.181
Merton	2.636	2.234
Heston	95.613	89.553
Heston-Merton	95.633	89.495
GARCH	4.411	3.525
GARCH-Merton	4.388	3.505

1-day · minutely rolling

Model	RMSE	MAE
GBM	2.800	2.294
Merton	2.785	2.270
Heston	94.324	87.693
Heston-Merton	93.434	86.626
GARCH	4.411	3.525
GARCH-Merton	4.387	3.503

5-day · hourly rolling

Model	RMSE	MAE
GBM	2.874	2.053
Merton	2.878	2.045
Heston	104.955	96.302
Heston-Merton	98.056	91.566
GARCH	4.616	3.636
GARCH-Merton	3.982	3.075

5-day · minutely rolling

Model	RMSE	MAE
GBM	3.118	2.121
Merton	3.223	2.193
Heston	100.493	90.787
Heston-Merton	97.652	88.472
GARCH	4.627	3.643
GARCH-Merton	4.007	3.118

Bold shaded row = lowest RMSE (LSM American call vs listed market). 12 contracts per ticker, 500 RN paths, seed 42. Heston-Merton is included. Mean is equal-weight across the

Option LSM vs market · SPY

1-day · hourly rolling

Model	RMSE	MAE
GBM	3.227	2.672
Merton	3.306	2.768
Heston	96.528	89.995
Heston-Merton	97.901	91.211
GARCH	4.257	3.002
GARCH-Merton	4.257	3.002

1-day · minutely rolling

Model	RMSE	MAE
GBM	3.425	2.799
Merton	3.382	2.765
Heston	104.550	96.164
Heston-Merton	104.051	95.226
GARCH	4.253	2.999
GARCH-Merton	4.243	2.986

5-day · hourly rolling

Model	RMSE	MAE
GBM	3.519	2.446
Merton	3.358	2.425
Heston	125.311	114.048
Heston-Merton	119.406	110.350
GARCH	4.605	3.328
GARCH-Merton	4.139	2.961

5-day · minutely rolling

Model	RMSE	MAE
GBM	4.024	2.619
Merton	4.060	2.678
Heston	122.963	108.615
Heston-Merton	121.327	106.650
GARCH	4.615	3.334
GARCH-Merton	4.597	3.268

Bold shaded row = lowest RMSE (LSM American call vs listed market). 12 contracts per ticker, 500 RN paths, seed 42. Heston-Merton is included.

Option LSM vs market · AAPL

1-day · hourly rolling

Model	RMSE	MAE
GBM	1.483	1.164
Merton	1.526	1.226
Heston	52.973	48.955
Heston-Merton	51.661	47.562
GARCH	1.649	1.332
GARCH-Merton	1.579	1.273

1-day · minutely rolling

Model	RMSE	MAE
GBM	1.685	1.224
Merton	1.678	1.177
Heston	53.729	49.372
Heston-Merton	51.823	47.031
GARCH	1.642	1.328
GARCH-Merton	1.581	1.278

5-day · hourly rolling

Model	RMSE	MAE
GBM	1.947	1.387
Merton	2.002	1.331
Heston	69.904	60.943
Heston-Merton	54.970	51.114
GARCH	2.478	2.042
GARCH-Merton	1.811	1.477

5-day · minutely rolling

Model	RMSE	MAE
GBM	2.008	1.342
Merton	2.114	1.382
Heston	68.071	57.969
Heston-Merton	51.341	47.720
GARCH	2.476	2.041
GARCH-Merton	1.802	1.487

Bold shaded row = lowest RMSE (LSM American call vs listed market). 12 contracts per ticker, 500 RN paths, seed 42. Heston-Merton is included.

Option LSM vs market · MSFT

1-day · hourly rolling

Model	RMSE	MAE
GBM	3.076	2.709
Merton	3.076	2.709
Heston	137.337	129.710
Heston-Merton	137.337	129.710
GARCH	7.327	6.241
GARCH-Merton	7.327	6.241

1-day · minutely rolling

Model	RMSE	MAE
GBM	3.289	2.858
Merton	3.295	2.869
Heston	124.692	117.543
Heston-Merton	124.429	117.621
GARCH	7.337	6.247
GARCH-Merton	7.337	6.246

5-day · hourly rolling

Model	RMSE	MAE
GBM	3.156	2.326
Merton	3.275	2.379
Heston	119.650	113.916
Heston-Merton	119.793	113.234
GARCH	6.766	5.536
GARCH-Merton	5.995	4.786

5-day · minutely rolling

Model	RMSE	MAE
GBM	3.322	2.401
Merton	3.494	2.519
Heston	110.446	105.777
Heston-Merton	120.287	111.045
GARCH	6.791	5.553
GARCH-Merton	5.622	4.599

Bold shaded row = lowest RMSE (LSM American call vs listed market). 12 contracts per ticker, 500 RN paths, seed 42. Heston-Merton is included.